



Unsupervised Learning



Supervised learning vs. unsupervised learning

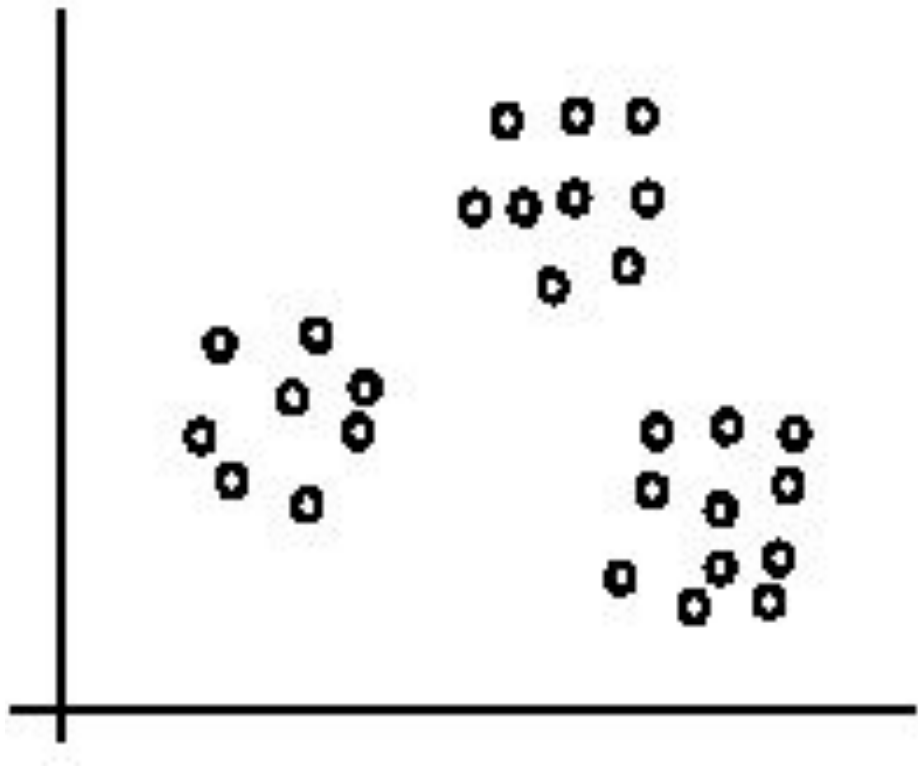
- **Supervised learning:** discover patterns in the data that relate data attributes with a target (class) attribute.
 - These patterns are then utilized to predict the values of the target attribute in future data instances.
- **Unsupervised learning:** The data have no target attribute.
 - We want to explore the data to find some intrinsic structures in them.

Clustering

- Clustering is a technique for finding **similarity groups** in data, called **clusters**. I.e.,
 - it groups data instances that are similar to (near) each other in one cluster and data instances that are very different (far away) from each other into different clusters.
- Clustering is often called an **unsupervised learning** task as no class values denoting an *a priori* grouping of the data instances are given, which is the case in supervised learning.
- Due to historical reasons, clustering is often considered synonymous with unsupervised learning.
 - In fact, association rule mining is also unsupervised
- This chapter focuses on clustering.

An illustration

- The data set has three natural groups of data points, i.e., 3 natural clusters.



What is clustering for?

- Let us see some real-life examples
- **Example 1:** groups people of similar sizes together to make “small”, “medium” and “large” T-Shirts.
 - Tailor-made for each person: too expensive
 - One-size-fits-all: does not fit all.
- **Example 2:** In marketing, segment customers according to their similarities
 - To do targeted marketing.

What is clustering for? (cont...)

- **Example 3:** Given a collection of text documents, we want to organize them according to their content similarities,
 - To produce a topic hierarchy
- **In fact, clustering is one of the most utilized data mining techniques.**
 - It has a long history, and used in almost every field, e.g., medicine, psychology, botany, sociology, biology, archeology, marketing, insurance, libraries, etc.
 - In recent years, due to the rapid increase of online documents, text clustering becomes important.

K-means clustering

- K-means is a **partitional clustering** algorithm
- Let the set of data points (or instances) D be $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$,
where $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ir})$ is a **vector** in a real-valued space $X \subseteq R^r$, and r is the number of attributes (dimensions) in the data.
- The k -means algorithm partitions the given data into k clusters.
 - Each cluster has a cluster **center**, called **centroid**.
 - k is specified by the user

K-means algorithm

- Given k , the *k-means* algorithm works as follows:
 - 1) Randomly choose k data points (**seeds**) to be the initial **centroids**, cluster centers
 - 2) Assign each data point to the closest **centroid**
 - 3) Re-compute the **centroids** using the current cluster memberships.
 - 4) If a convergence criterion is not met, go to 2).

K-means algorithm – (cont ...)

Algorithm k -means(k, D)

```
1  Choose  $k$  data points as the initial centroids (cluster centers)
2  repeat
3      for each data point  $\mathbf{x} \in D$  do
4          compute the distance from  $\mathbf{x}$  to each centroid;
5          assign  $\mathbf{x}$  to the closest centroid      // a centroid represents a cluster
6      endfor
7      re-compute the centroids using the current cluster memberships
8  until the stopping criterion is met
```

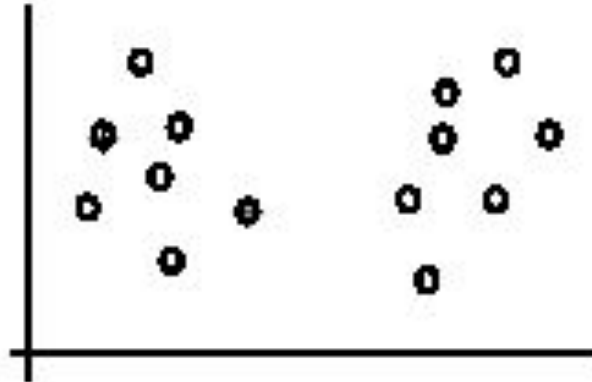
Stopping/convergence criterion

1. no (or minimum) re-assignments of data points to different clusters,
2. no (or minimum) change of centroids, or
3. minimum decrease in the **sum of squared error (SSE)**,

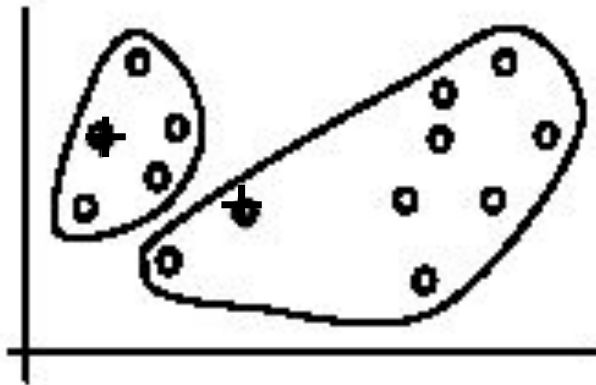
$$SSE = \sum_{j=1}^k \sum_{\mathbf{x} \in C_j} \text{dist}(\mathbf{x}, \mathbf{m}_j)^2 \quad (1)$$

- C_j is the j th cluster, $\mathbf{m}_j$ is the centroid of cluster C_j (the mean vector of all the data points in C_j), and $\text{dist}(\mathbf{x}, \mathbf{m}_j)$ is the distance between data point $\mathbf{x}$ and centroid $\mathbf{m}_j$.

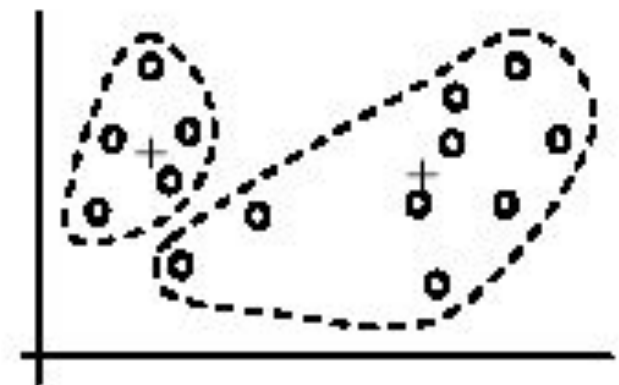
An example



(A). Random selection of k centers

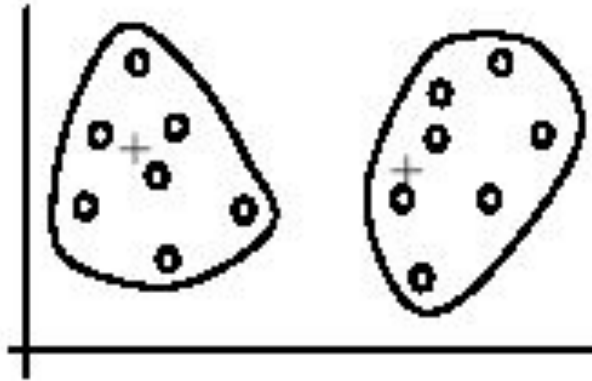


Iteration 1: (B). Cluster assignment

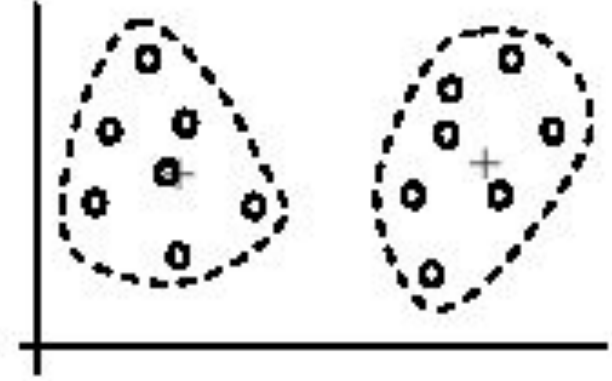


(C). Re-compute centroids

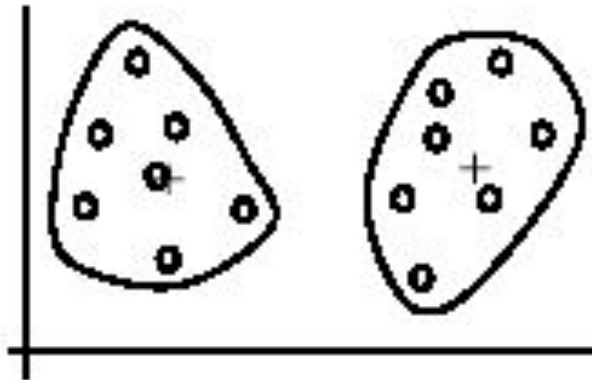
An example (cont ...)



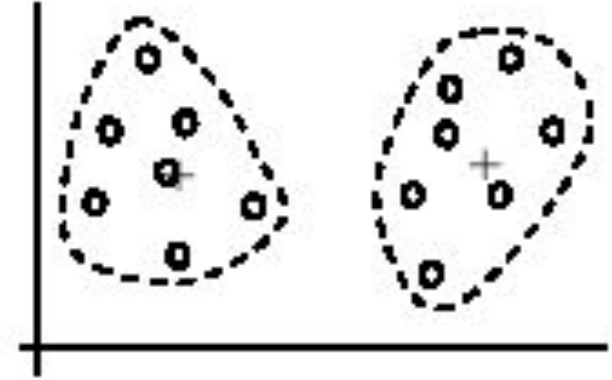
Iteration 2: (D). Cluster assignment



(E). Re-compute centroids



Iteration 3: (F). Cluster assignment



(G). Re-compute centroids

An example distance function

The k -means algorithm can be used for any application data set where the **mean** can be defined and computed. In the **Euclidean space**, the mean of a cluster is computed with:

$$\mathbf{m}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i \quad (2)$$

where $|C_j|$ is the number of data points in cluster C_j . The distance from one data point $\mathbf{x}_i$ to a mean (centroid) $\mathbf{m}_j$ is computed with

$$\begin{aligned} \text{dist}(\mathbf{x}_i, \mathbf{m}_j) &= \|\mathbf{x}_i - \mathbf{m}_j\| \\ &= \sqrt{(x_{i1} - m_{j1})^2 + (x_{i2} - m_{j2})^2 + \dots + (x_{ir} - m_{jr})^2} \end{aligned} \quad (3)$$

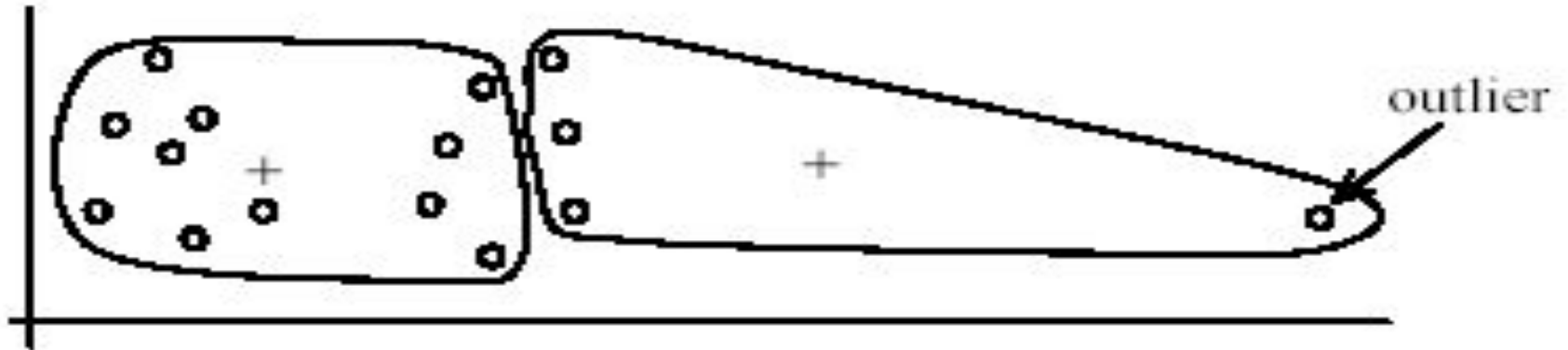
Strengths of k-means

- Strengths:
 - Simple: easy to understand and to implement
 - Efficient: Time complexity: $O(tkn)$,
where n is the number of data points,
 k is the number of clusters, and
 t is the number of iterations.
 - Since both k and t are small. k -means is considered a linear algorithm.
- K-means is the most popular clustering algorithm.
- Note that: it terminates at a **local optimum** if SSE is used. The **global optimum** is hard to find due to complexity.

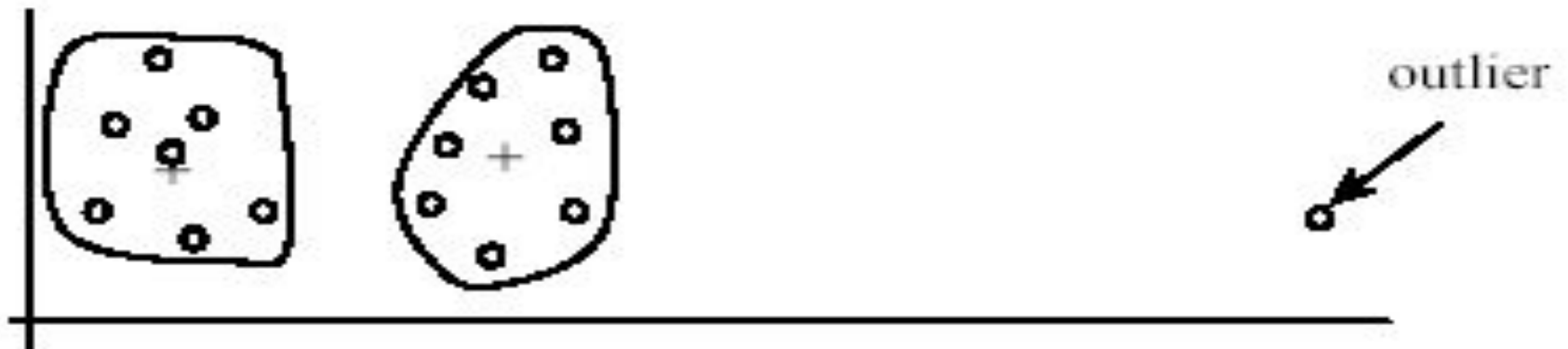
Weaknesses of k-means

- The algorithm is only applicable if the **mean** is defined.
 - For categorical data, *k*-mode - the centroid is represented by most frequent values.
- The user needs to specify ***k***.
- The algorithm is sensitive to **outliers**
 - Outliers are data points that are very far away from other data points.
 - Outliers could be errors in the data recording or some special data points with very different values.

Weaknesses of k-means: Problems with outliers



(A): Undesirable clusters



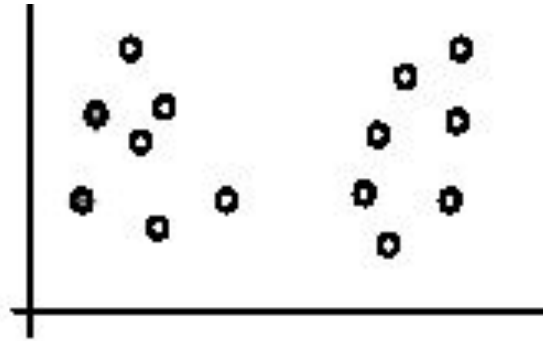
(B): Ideal clusters

Weaknesses of k-means: To deal with outliers

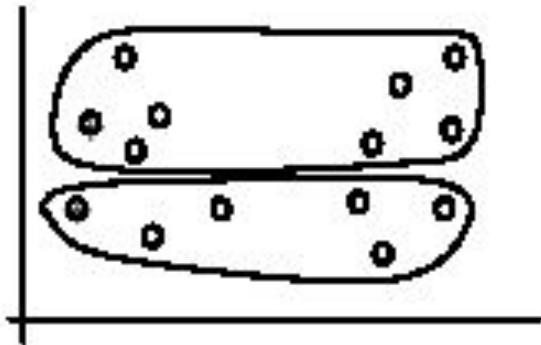
- One method is to remove some data points in the clustering process that are much further away from the centroids than other data points.
 - To be safe, we may want to monitor these possible outliers over a few iterations and then decide to remove them.
- Another method is to perform random sampling. Since in sampling we only choose a small subset of the data points, the chance of selecting an outlier is very small.
 - Assign the rest of the data points to the clusters by distance or similarity comparison, or classification

Weaknesses of k-means (cont ...)

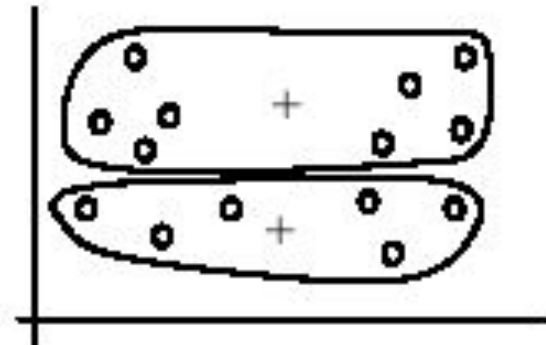
- The algorithm is sensitive to **initial seeds**.



(A). Random selection of seeds (centroids)



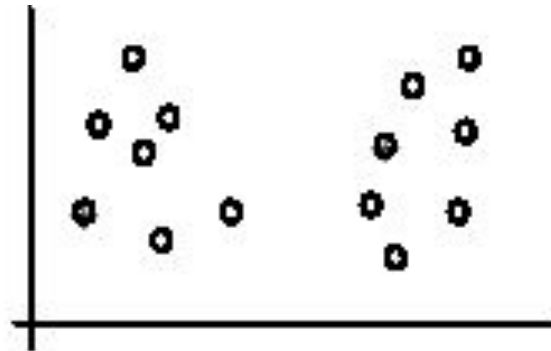
(B). Iteration 1



(C). Iteration 2

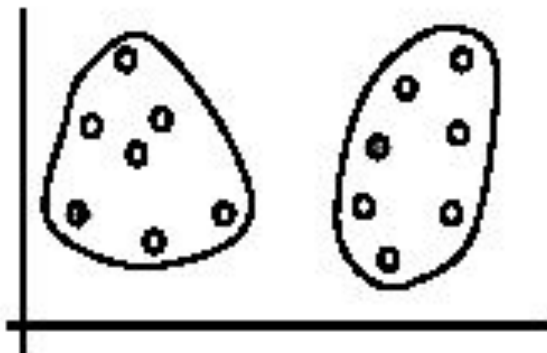
Weaknesses of k-means (cont ...)

- If we use **different seeds**: good results

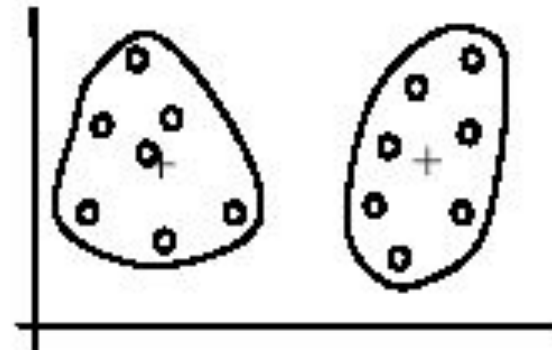


(A). Random selection of k seeds (centroids)

- There are some methods to help choose good seeds



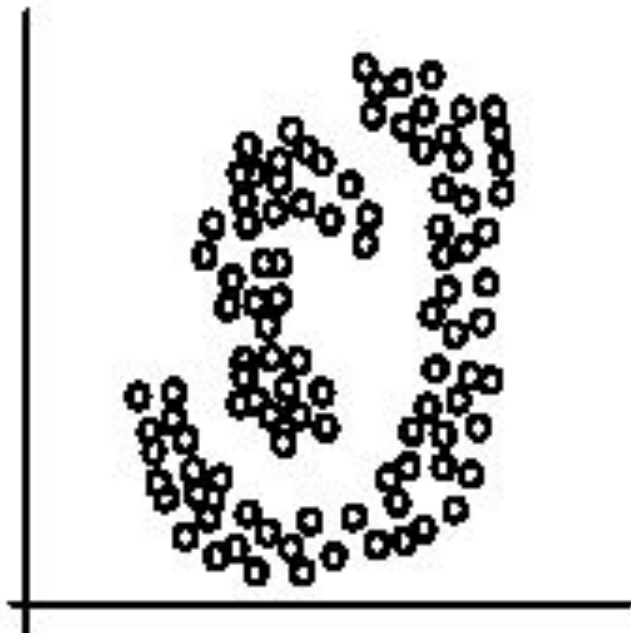
(B). Iteration 1



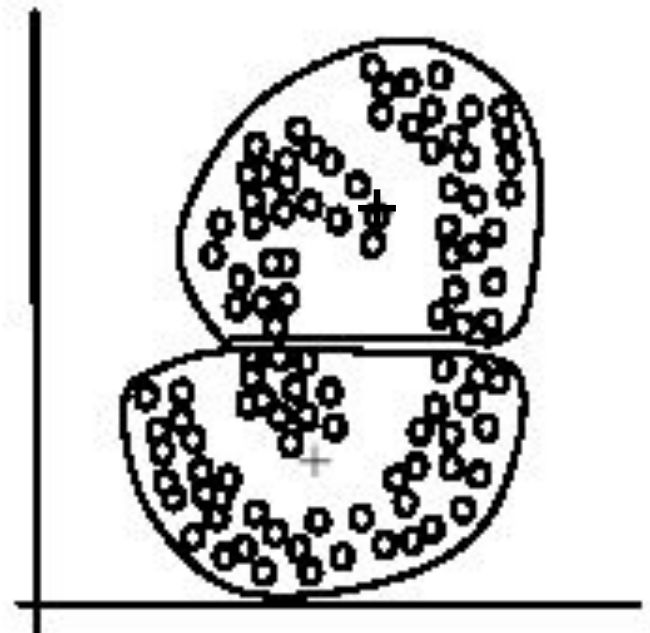
(C). Iteration 2

Weaknesses of k-means (cont ...)

- The k -means algorithm is not suitable for discovering clusters that are not hyper-ellipsoids (or hyper-spheres).



(A): Two natural clusters



(B): k -means clusters

K-means summary

- Despite weaknesses, *k*-means is still the most popular algorithm due to its simplicity, efficiency and
 - other clustering algorithms have their own lists of weaknesses.
- No clear evidence that any other clustering algorithm performs better in general
 - although they may be more suitable for some specific types of data or applications.
- Comparing different clustering algorithms is a difficult task. No one knows the correct clusters!